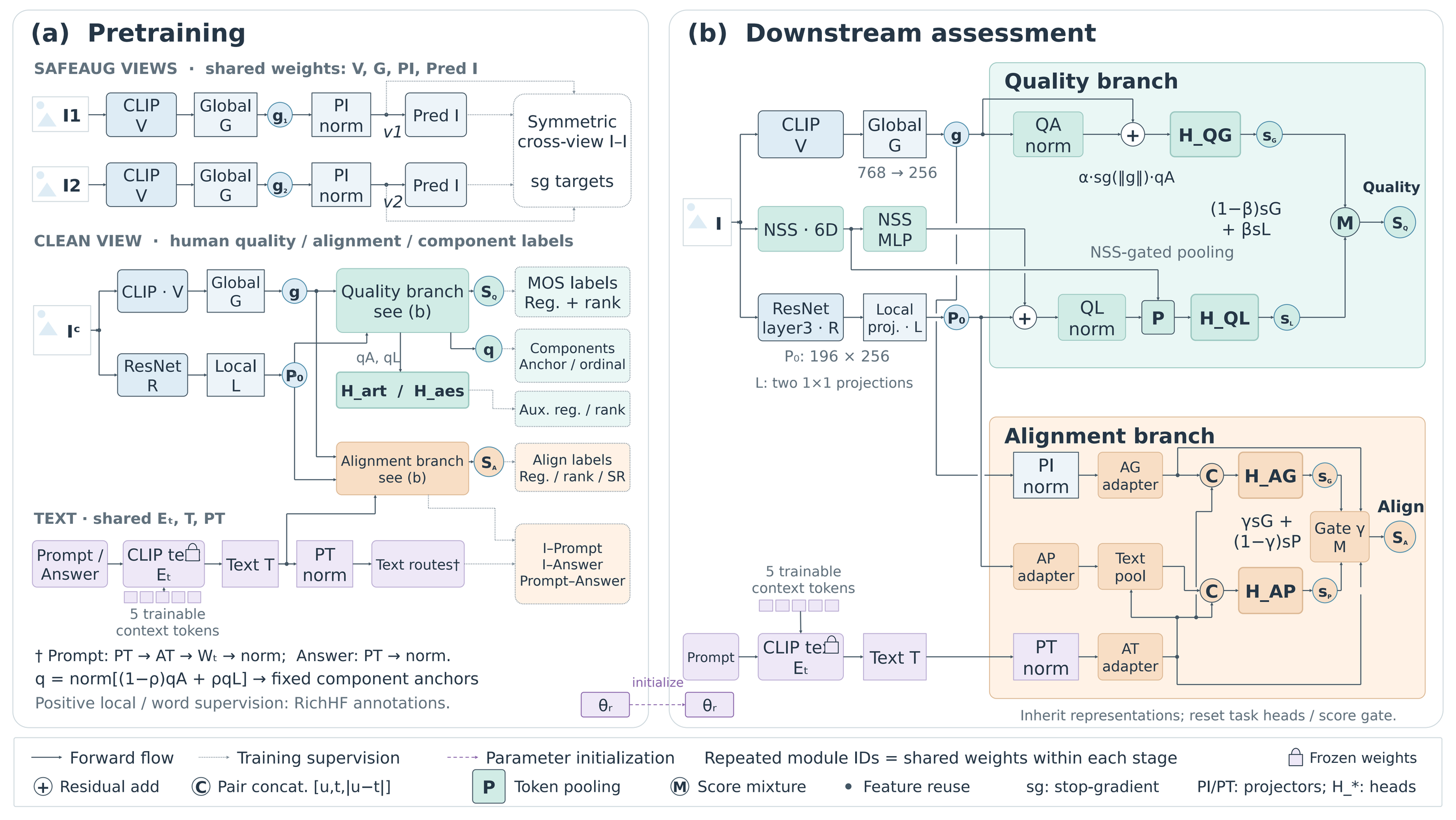

(a) Pretraining
SAFEAUG VIEWS · shared weights: V, G, PI, Pred I
I1
CLIP V
Global G
g1
PI norm
v1
Pred I
I2
CLIP V
Global G
g2
PI norm
v2
Pred I
Symmetric cross-view I–I
sg targets
CLEAN VIEW · human quality / alignment / component labels
Ic
CLIP · V
Global G
g
ResNet R
Local L
P0
Quality branch see (b)
So
MOS labels Reg. + rank
q
Components Anchor / ordinal
qA, qL
H_art / H_aes
Aux. reg. / rank
Alignment branch see (b)
SA
Align labels Reg. / rank / SR
TEXT · shared Et, T, PT
Prompt / Answer
CLIP te Et
5 trainable context tokens
Text T
PT norm
Text routes†
I–Prompt
I–Answer
Prompt–Answer
† Prompt: PT → AT → Wt → norm; Answer: PT → norm.
q = norm[(1−ρ)qA + ρqL] → fixed component anchors
Positive local / word supervision: RichHF annotations.
initialize
θr
θr
(b) Downstream assessment
Quality branch
I
CLIP V
Global G
768 → 256
g
QA norm
H_QG
sG
α·sg(‖g‖)·qA
Quality
(1−β)sG + βsL
M
So
NSS-gated pooling
NSS · 6D
NSS MLP
ResNet layer3 · R
Local proj. · L
P0
P0: 196 × 256
L: two 1×1 projections
QL norm
P
H_QL
sL
Alignment branch
PI norm
AG adapter
C
H_AG
sG
γsG + (1−γ)sP
Gate γ M
Align
SA
AP adapter
Text pool
C
H_AP
sP
5 trainable context tokens
Prompt
CLIP te Et
Text T
PT norm
AT adapter
Inherit representations; reset task heads / score gate.
Forward flow
Training supervision
Parameter initialization
Repeated module IDs = shared weights within each stage
Frozen weights
Residual add
Pair concat. [u,t,|u−t|]
P Token pooling
M Score mixture
Feature reuse
sg: stop-gradient
PI/PT: projectors; H_*: heads